MIL-C-48411/28A(MU)
1 December 1975
SUPERSEDING
MIL-C-48411/28(MU)
1 October 1974

MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 117465258
MEMORY II

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Power Control.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

117465258	Circuit Card Assembly, Memory II

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. *REQUIREMENTS*

3.1 Item identification.- This assembly is second of two single ended printed circuit boards consisting 512 bits, Bi-Polar programmable read - only - memories which supply the digital information for 100 programed flights for the Simulated Target Generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745258.

3.2 Characteristics.

FSC 1240

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with a + 5VDC Power Supply.

3.2.1.2 Input/Output Test.- When specific logic inputs are applied a corresponding output shall result as in Table I.

3.2.2 Environmental.- The assembly shall maet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745258 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 117465258and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C 48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power Supply.- The power supply shall be 10.00 + 0.05 VDC, 100 mA or greater, an a ripple of less than 10 MV RMS

4.2.1.2 Dummy Loads.- The dummy load shall be eight triple 3-input positive nand gates type SN5410 or equilvalent.

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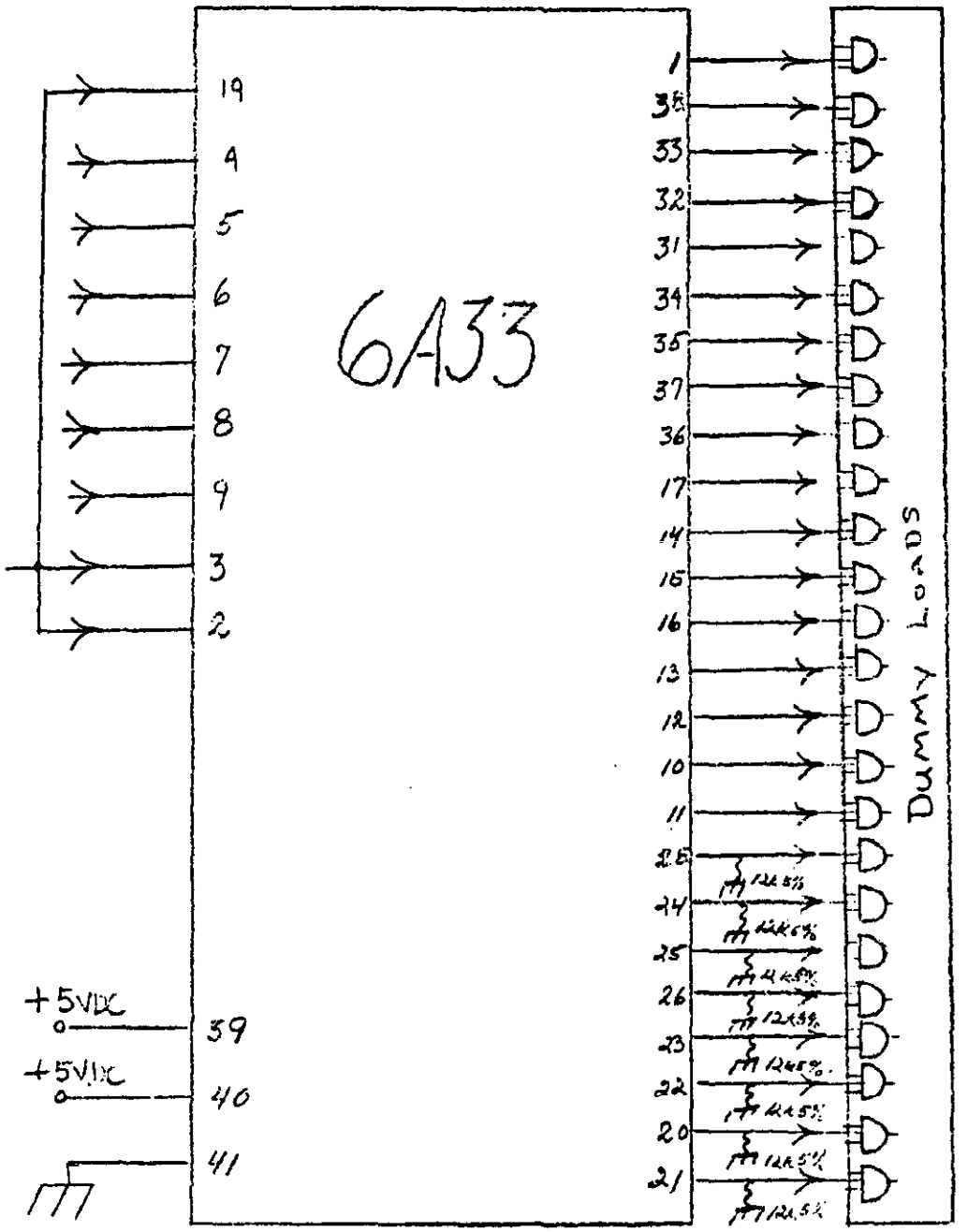
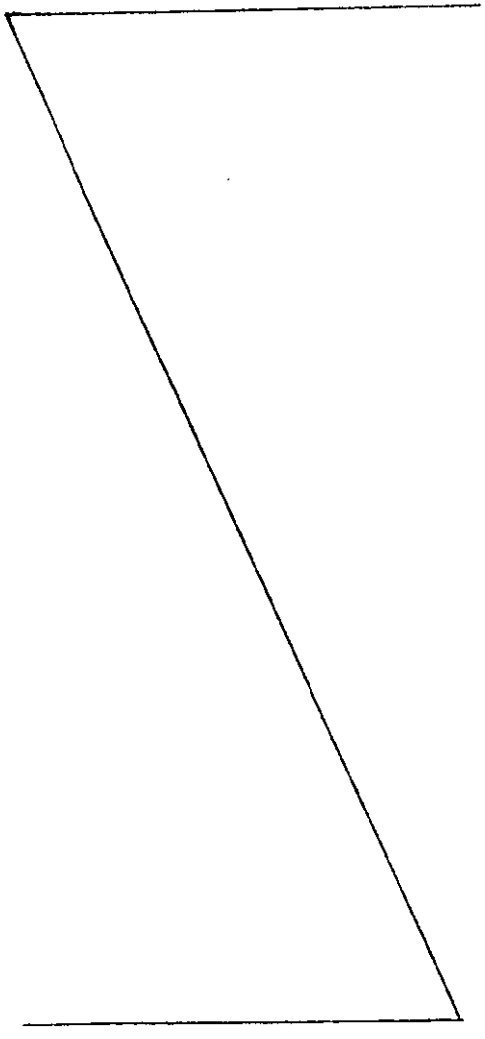


FIGURE 1

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TABLE I - INPUT/OUTPUT MEMORY II TEST

	INPUT							OUTPUT																					
Dec.	3	9	8	7	6	5	4	11	10	12	13	16	15	14	17	36	37	35	34	31	32	33	38	21	20	22	23	26	25	24	28
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
3	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0												
4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
5	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
6	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
7	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0												
8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
9	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
10	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
11	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0												
12	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
13	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
14	0	0	0	1	1	1	0	0	0	0	0	1	1	1	0	0	0	0	0												
15	0	0	0	1	1	1	1	0	0	0	1	0	0	0	1	1	1	1	1												
16	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	1	1	1	1												
17	0	0	1	0	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0												
18	0	0	1	0	0	1	0	0	0	0	1	1	0	1	0	0	0	0	0												



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TABLE I - INPUT/OUTPUT MEMORY II TEST

Dec.	INPUT 3	9	8	7	6	5	4	OUTPUT 11	10	12	13	16	15	14	17	36	37	35	34	31	32	33	38	21	20	22	23	26	25	24	28
19	0	0	1	0	0	1	1	1	0	1	0	0	0	0	0	0	0	0	0												
20	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0												
21	0	0	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0												
22	0	0	1	0	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0												
23	0	0	1	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0												
24	0	0	1	1	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0												
25	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0												
26	0	0	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0												
27	0	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0												
28	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0												
29	0	0	1	1	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0												
30	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
31	0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0												
32	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
33	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
34	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
35	0	1	0	0	0	1	1	0	0	0	0	0	1	0	1	1	1	1	1												
36	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
37	0	1	0	0	1	0	1	0	0	0	0	1	0	0	1	0	0	0	0												

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TABLE I - INPUT/OUTPUT MEMORY II TEST

	INPUT							OUTPUT																							
Dec.	3	9	8	7	6	5	4	11	10	12	13	16	15	14	17	36	37	35	34	31	32	33	38	21	20	22	23	26	25	24	28
38	0	1	0	0	1	1	0	0	0	0	0	1	0	0	1	0	1	0	1												
39	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0												
40	0	1	0	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0												
41	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
42	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
43	0	1	0	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0												
44	0	1	0	1	1	0	0	0	1	1	0	0	1	0	1	1	0	0	0												
45	0	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
46	0	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
47	0	1	0	1	1	1	1	0	1	1	0	1	0	0	1	1	0	0	0												
48	0	1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0												
49	0	1	1	0	0	0	1	0	1	1	1	0	0	1	0	0	0	0	0												
50	0	1	1	0	0	1	0	0	1	1	1	1	0	1	0	1	0	0	0												
51	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0												
52	0	1	1	0	1	0	0	0	1	1	0	0	1	0	1	0	0	0	0												
53	0	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0												
54	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0												
55	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0												

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TABLE I - INPUT/OUTPUT MEMORY II TEST

Dec.	INPUT 3	9	8	7	6	5	4	11	10	12	13	16	15	14	17	36	37	35	34	OUTPUT 31	32	33	38	21	20	22	23	26	25	24	28
56	0	1	1	1	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1												
57	0	1	1	1	0	0	1	0	1	1	0	0	1	0	0	1	1	1	0												
58	0	1	1	1	0	1	0	1	0	0	0	0	1	0	0	1	1	0	0												
59	0	1	1	1	0	1	1	0	1	0	0	0	1	0	1	0	1	0	1												
60	0	1	1	1	1	0	0	0	1	0	0	1	1	0	1	1	0	0	0												
61	0	1	1	1	1	0	1	0	1	0	0	1	1	0	1	0	0	0	0												
62	0	1	1	1	1	1	0	0	0	0	0	0	1	0	1	0	1	0	1												
63	0	1	1	1	1	1	1	0	0	0	0	1	0	0	1	0	0	0	0												
64	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	0	0					0	1	1	1	1	0	1	1
65	1	0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	1	1	0					1	0	0	1	1	0	0	1
66	1	0	0	0	0	1	0	0	1	1	1	0	0	0	1	1	0	0	0					1	1	0	1	1	1	0	0
67	1	0	0	0	0	1	1	0	1	0	0	1	1	0	1	0	0	0	0					1	1	0	1	1	0	0	1
68	1	0	0	0	1	0	0	0	1	1	1	1	0	1	0	0	0	0	0					1	1	1	1	0	1	0	0
69	1	0	0	0	1	0	1	0	1	0	1	0	0	0	1	1	0	0	0					1	1	1	1	0	0	0	1
70	1	0	0	0	1	1	0	0	1	0	0	0	1	0	1	0	1	0	1					0	1	1	1	1	0	1	1
71	1	0	0	0	1	1	1	0	1	0	0	0	1	0	1	0	1	0	1					1	1	1	1	1	0	1	1
72	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0
73	1	0	0	1	0	0	1	0	1	0	1	0	1	1	0	0	0	0	0					1	1	1	1	0	0	1	0
74	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0	0					1	1	1	1	0	0	0	1

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TABLE I - INPUT/OUTPUT MEMORY II TEST

Dec.	INPUT							OUTPUT																				
	3	9	8	7	6	5	4	11	10	12	13	16	15	14	17	36	37	35	34	31	32	33	38	21	20	22	23	26	25	24	28
75	1	0	0	1	0	1	1	0	1	1	0	0	1	0	1	0	1	0	1					1	1	1	1	0	1	0	1
76	1	0	0	1	1	0	0	0	1	0	0	1	0	0	1	0	1	0	1					0	1	1	1	0	0	1	1
77	1	0	0	1	1	0	1	0	1	0	0	0	1	0	0	1	1	0	0					1	1	1	1	0	0	1	0
78	1	0	0	1	1	1	0	1	0	0	0	0	1	0	1	0	1	1	0					1	1	1	1	0	1	0	1
79	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0
80	1	0	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0					1	1	0	0	0	0	1	1
81	1	0	1	0	0	0	1	0	1	1	1	1	0	1	0	0	0	0	0					1	1	1	1	0	0	1	1
82	1	0	1	0	0	1	0	0	1	0	1	0	1	0	1	1	0	0	0					1	1	1	1	0	0	0	1
83	1	0	1	0	0	1	1	0	1	0	0	0	1	0	0	1	0	1	1					1	1	1	1	0	0	1	1
84	1	0	1	0	1	0	0	0	1	0	0	1	0	0	1	0	0	0	0					1	1	1	1	0	0	1	0
85	1	0	1	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0					1	1	0	0	0	0	1	0
86	1	0	1	0	1	1	0	0	1	1	0	1	1	0	1	0	1	0	1					1	1	1	1	0	1	0	1
87	1	0	1	0	1	1	1	0	1	0	0	1	0	0	1	1	0	0	0					1	1	0	0	0	0	1	0
88	1	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0					1	1	0	0	0	0	0	1
89	1	0	1	1	0	0	1	0	1	0	0	0	1	0	1	0	1	0	1					1	0	0	1	0	0	1	1
90	1	0	1	1	0	1	0	0	1	0	0	1	1	0	1	1	0	0	0					1	0	0	1	0	0	1	1
91	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	0	0	0	0					1	0	0	1	0	0	1	0
92	1	0	1	1	1	0	0	0	1	0	0	0	1	0	1	1	0	0	0					1	0	1	1	0	0	1	1

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TABLE I - INPUT/OUTPUT MEMORY II TEST

	INPUT							OUTPUT																				
Dec.	3	9	8	7	6	5	4	11	10	12	13	16	15	14	17	36	37	35	34	31	32	33	38	21	20	22	23	26	25	24	28
93	1	0	1	1	1	0	1	0	1	1	1	0	0	0	1	1	0	0	0					0	0	0	1	1	1	0	0
94	1	0	1	1	1	1	0	0	1	0	1	0	0	0	1	1	0	0	0					0	0	1	1	0	0	0	1
95	1	0	1	1	1	1	1	0	1	0	0	0	1	0	1	0	1	0	1					1	0	1	1	1	0	1	1
96	1	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	0	1					0	0	1	1	1	0	1	1
97	1	1	0	0	0	0	1	1	0	0	0	1	0	0	1	0	0	0	0					0	0	1	1	1	1	1	0
98	1	1	0	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0	0					0	0	1	1	1	0	1	1
99	1	1	0	0	0	1	1	0	1	1	0	0	1	0	1	0	1	1	0					1	1	1	1	0	0	1	1

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4.2.1.3 Digital voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All the requirements shall be met with a power supply of + 5.00 ± 0.05 VDC.

4.2.2.2 Digital output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between + 2.40 and 5.00 VDC.

4.2.2.3 Test set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Input/Output Test.- Connect the Circuit Card as shown in Figure I. When digital "1"s, and "0"s, as defined in paragraph 4.2.2.2 above, are applied to the respective input pins of Table 1 the corresponding output shall be as shown.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745258.

6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A664

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DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-C-48411/28A(MU) Circuit Card Assembly: 117465258 Memory II

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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